

**Prenatal Viral Infection and its Impact on Rodent Heart Rate Variability (HRV); Start of
an Exploratory Journey into Offspring Cognitive Flexibility**

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Abstract

This study aimed to establish a relationship between prenatal infections and changes in HRV between offspring exposed to viral infections in utero vs those that were not. Prenatal viral infections were induced in Sprague Dawley rats via injection of saline , 2mg/kg, or 5mg/kg PolyI:C. Pups' heart rates were measured on postnatal day (PND) 7 and PND14 and transformed into six HRV parameters. We hypothesized that pups exposed to PolyI:C vs a saline solution, during gestation, will have decreased HRV, whereas pups whose mother was administered the saline solution will see a higher HRV than the PolyI:C pups. This was partially confirmed by a significant PND and dose interaction within the HF (nu) HRV parameter, Post Hoc analyses confirmed the difference to lie between 2mg/kg and 5mg/kg PolyI:C doses, with the 2mg/kg dose producing higher HF (nu) values. This meant that pups in the 2mg/kg dose condition were experiencing higher PNS activity than those in the 5mg/kg dose condition, meaning that the pups' HRV of the lower dose of PolyI:C was affected more than those who experienced the higher dose. Taking a look at the tendencies between PND and dose means, a similar conclusion was drawn.

Introduction

Heart rate variability (HRV) is the variation of time between each heartbeat (Perna et. al., 2020). HRV is measured by observing the decomposition of 'RR' intervals from the electrical activity that is picked up by an electrocardiogram (ECG). An RR interval is the time between two R-waves (Lanfranchi & Somers, 2010). An R-wave is the reflection and measure of the depolarization of the main mass of the heart's ventricles (Ashley & Niebauer, 2004). HRV is a product of the bidirectional communication between the brain and the heart. The inhibitory network that is in close communication with the heart and influences HRV is the central autonomic network (CAN). CAN is responsible for the regulation of the sympathetic (SNS) and parasympathetic nervous system (PNS), which plays a role in the response to external stimuli and partially determines the response one may have to e.g., a bear. CAN is comprised of subcortical structures that aid in functioning: prefrontal cortex, cingulate cortex, insula, amygdala, and regions in the brain stem (Thayer et al., 2009; Thayer & Lane, 2000, 2009). Signals to mediate HRV travel from preganglionic (cells supplying signals to ganglions) sympathetic and parasympathetic neurons down to the stellate ganglia in the neck and through the vagus nerve to the heart and back. In turn, the SNS and PNS are mediated by the neurotransmitters norepinephrine and acetylcholine respectively. Acetylcholine works to inhibit the heart muscles and slow down heart rate, while norepinephrine works to excite heart muscles which speeds up heart rate. HRV is mostly influenced by the parasympathetic part of the nervous system, meaning that acetylcholine is one of the primary neurotransmitters involved in the regulation of HRV (Kemp & Quintana, 2013). The communication between brain and heart can also be influenced by other physiological factors, outside the aforementioned subcortical

structures, such as thermoregulation, renin-angiotensin system, hormones, etc. (Perna et. al., 2020).

HRV has been identified as a measure of risk assessment for (future) heart-related diseases but also as an indicator of emotional and cognitive inflexibility which is in turn linked to the development/existence of emotional regulation issues implicated in mental disorders in the individual (Zhu et. al., 2019). Upon sudden exposure to a potentially harmful stimulus, the SNS will prepare for the next mode of action, 'fight or flight'. Variation in HRV has been shown to be reflective of beneficial short- and long-term reactions to a sudden threat and is seen as a display of greater emotion regulation/flexibility. Therefore, the opposite, emotional inflexibility, has been linked to a decreased efficiency in identifying and responding to potentially threatening stimuli by numerous clinical studies and meta-analyses. Calkins and Keane (2004) found in their sample of preschool children that lower HRV indicated children would struggle more socially and emotionally than children with higher HRV. Furthermore, Miskovic and Schmidt (2010) pointed to an increased attentional bias toward negative affective stimuli. Not only do individuals with low HRV struggle with emotional flexibility, but they also tend to selectively focus on negative stimuli more than individuals with high HRV (Park et. al., 2013). This means that they are seeking out negativity without being able to adaptively cope with it. Moreover, low HRV has also been linked to worse cognitive outcomes for individuals of all ages. Hansen et. al. (2003) conducted a study measuring cognitive ability with military personnel and found there to be a disparity in executive function and working memory between low and high HRV participants, with high HRV participants producing higher scores. When the same participants were put under stress, only the low HRV group improved on their previous scores and it was thought this was because the high HRV group had already reached their full potential. In a study with elderly

participants, Mahinrad et. al., (2016) found there to be a significant difference in reaction time between low and high HRV participants, with low HRV producing slower reaction times than those with high HRV. They also found that these results could have been predicted with the 10-second-long HRV measurements taken prior to testing, which matched up with the participants' task performance.

It is becoming evident from these studies with participants varying in tasks administered, age, occupation, etc., that HRV can be a useful predictor for emotional and cognitive flexibility. With this knowledge we could try to predict the occurrence of mental disorders, as mental disorders are usually associated with problems in the emotional and/or cognitive domain. Such measurements have been done and were summarised in Ellenbroek et. al.'s (2019) review. They discussed HRV's possible role in depression, anxiety, schizophrenia, autism spectrum disorder, bipolar disorders, addiction, and bulimia. In their paper, a link was drawn between the areas in the brain involved in both HRV and the major mental disorders, the main areas discussed being the prefrontal cortex, anterior cingulate, insula, amygdala, and striatum. Though this involvement of similar parts of the brain in HRV and mental disorders sounds exciting by itself, it does not confirm a direct relationship between HRV and mental disorders as other processes also require similar areas. This observation should be handled as one point of evidence of HRV's possible involvement in mental health outcomes. However, when taking a closer look at the different mental disorders the picture becomes clearer, a decrease in HRV was present in people affected by MDD, schizophrenia, ASD, GAD, and during depressive episodes in bipolar disorder (with an increase in HRV found during manic episodes). BuliBulimia-affectedple, on the other hand, were found to have increased HRV (Ellenbroek et. al., 2019).

There are many different parameters/measures within HRV that we can extract from heart rate data that can inform us of activity within the PNS and SNS. There are linear and non-linear measurements, in this study, we will be focusing on linear measurements of time and frequency. RMSSD, the root mean square of successive differences, is a linear measurement of time. It is the most commonly used parameter as it mainly assesses PNS activity which is most relevant as PNS activity is primarily impacted in individuals with neurodevelopmental disorders (Ellenbroek et al., 2019). High Frequency (HF), HF (%), HF normalised units (HF (nu)), low frequency (LF), LF (%), LF normalised units (LF (nu)), and LF/HF are all linear measurements of frequency (Tiwari et al., 2022). HF reflects PNS activity and is a respiratory band range between 0.15Hz and 0.4Hz. HF (%) is the percentage of HF frequency band in the total power spectrum (TP). TP is the total variance and is the same as the sum of the spectral bands LF, HF, very low frequency (VLF), and ultra low frequency (ULF) (Kidwell & Ellenbroek, 2018; Medicore, 1996). TP reflects the overall activity of the autonomic nervous system (ANS) and lies in the range of 0 to 0.4Hz (Medicore, 1996; Tiwari et al., 2022). However, you cannot distinguish between PNS and SNS activity within the ANS using solely TP (Kidwell & Ellenbroek, 2018). Therefore, we extract measures like HF and LF from TP to make those distinctions. LF reflects mainly SNS activity with some contributions from PNS and is a power band spectrum ranging between 0.04 and 0.15Hz. LF (%) is the percentage of LF frequency band in the TP (Tiwari et al., 2022). HF (nu) and LF (nu) are both ‘normalised’ versions of HF and LF. For the measures to be normalised we apply these equations, as mentioned in Tiwari et al. (2022), respectively: $[HF / (Total\ Power - VLF)] \times 100$ and $[LF / (Total\ Power - VLF)] \times 100$. The normalization of HF and LF aids in highlighting the intricate balance between the PNS and SNS in the ANS, also tends to reduce noise as VLF is excluded from TP which aids in focusing solely on HF and LF

(Electrophysiology, 1966). The last linear measure we are including in this study is LF/HF, which you attain by calculating the ratio between LF and HF. This measure is a reflection of the balance between the PNS and SNS. Higher LF/HF indicates the dominance of the SNS over the PNS and lower values indicate the reverse (Kidwell & Ellenbroek, 2018; Tiwari et al., 2022).

Viral and bacterial infections during pregnancy have been connected to the development of mental disorders in the offspring. In our study we will focus on viral infections. During a viral infection, inflammation increases throughout the pregnant person's body in a bid to fight the virus. This inflammation has an impact on the placenta's ability to help the child develop properly. Sometimes maternal infection may also lead to the development of mental disorders in the offspring. Massrali et al. (2022) reviewed human epidemiological data from numerous studies that were concerned with the increased likelihood of the offspring experiencing a neurodevelopmental disorder (autism, schizophrenia, ADHD, and depression) when the mother was exposed to viruses like influenza during pregnancy. From those reviews, they concluded that maternal viral infections were likely contributing to the increase in neurodevelopmental disorders. Penner and Brown (2007) conducted viral maternal infection studies, specifically looking at the incidence of schizophrenia when the mother was exposed to influenza. They found that the likelihood of developing schizophrenia increased by seven times when the mother had been exposed to influenza during pregnancy.

As mentioned earlier, low HRV has been found in individuals with autism, schizophrenia, depression, ADHD, anxiety, and depressive episodes in bipolar (with an increase in HRV in manic episodes and also in individuals affected by bulimia) (Ellenbroek et al., 2019). However, a link has not yet been made between maternal viral infections and HRV. Our study is aiming to fill this gap in the literature by focusing on the potential effect a synthetically induced

maternal viral infection can have on the offspring's HRV. From the results we can infer, based on the number of offspring with low HRV when exposed to a viral infection prenatally, if maternal viral infections have an effect on HRV. If there is an effect, it is important to consider HRV as a measure for possible future mental disorders along with the knowledge of past environmental influences such as prenatal infections.

To solidify HRV as a reliable measure of possible current or future mental disorders we conducted our study. By adding to the current literature on using HRV as an indicator of possible mental illness, we could help inspire further research into the matter which could lead to the normalisation of the use of HRV as a diagnostic tool in the world of medicine and psychology, ultimately optimising the treatment process for many people affected by their mental disorders. Our study aims to establish a relationship between prenatal infections and changes in HRV between offspring exposed to viral infections in utero vs those that were not. We are choosing to use an animal model, utilising Sprague Dawley rats, to measure this relationship. An animal model has its obvious limitations in regard to how results produced in this study could be mapped onto the human experience. However, animal models have the advantage of control. We can control every aspect of the rat mothers' lives and are able to manipulate the variable we are interested in. We hypothesise that pups exposed to PolyI:C vs a saline solution, during gestation, will have decreased HRV, whereas pups whose mother was administered the saline solution will see a higher HRV than the PolyI:C pups.

We will inject pregnant rats on gestational day 12 with PolyI:C, a substance that mimics a viral infection. PolyI:C has been used as a reliable substance to induce viral-infection-like symptomology in many studies and has shown great face, construct, and predictive validity (Haddad et. al., 2020; Missault et al., 2019; Patterson, 2011). Once the pups are born, we will

measure their HRV on post-natal days 7 and 14. We will also pay close attention to possible sex differences.

Method

Design

We applied an experimental post-test control group design. The independent variables were sex (female/male) and dosage of PolyI:C/saline (2mg/kg or 5mg/kg of PolyI:C, saline solution), these two variables were both classed as being between-groups. The dependent variables were parameters of HRV: RMSSD, LF (%), LF (nu), HF (%), HF (nu), and LF/HF.

Animals

We used Sprague Dawley rats for our experiment. The rats were all housed at the Te Toki a Rata vivarium at Victoria University of Wellington. The cage room was kept at a constant temperature of $20^{\circ} \pm 2^{\circ}$ and the rats lived a reversed 12-hour light/dark cycle (active hours when measured being 10am to 4pm). Two rat pups (one female and one male) were randomly picked out of each litter to reduce litter effects.

Materials

We used ECGenie to measure heart rate. First we ensured that the pad responsible for conducting the pups' HRV was warm to the touch to be comfortable for the pup (around 25°C to 32°C , as recommended by Melissa Lewis at the Mouse Specifics department at ECGenie). Once this was achieved, we spread conductive gel onto the front and back of the pad to increase conductivity for ideal readings. The pup was then brought in and placed on the pad, then a small metal half-cage was secured over the pup to ensure minimal movement, which could lead to

disturbance in the signal. The readings were visualized, observed, and edited using Lab-Chart 8 and if anything appeared unusual and lasted for too long, the pup was removed, the pad re-gelled, and the pup was placed back on the pad, usually this would resolve the issue. After measurement, the pup was returned to the litter, the pad was cleaned, and the process was repeated with the remaining pup in the initial cage and then all the other pups from the other cages. My lab partner's focus was the ultrasonic vocalizations (USVs). While the pups' HRV was being measured, a microphone was holstered above the ECGenie to pick up the pups' squeaks and USVs. The USVs were recorded using the program "Audacity".

Procedure

Male and female rats were paired by placing them in the same cage which had a metal grate with holes inserted to allow droppings and vaginal plugs to fall through. The pairs' cages were scanned in the morning and afternoon for vaginal plugs until at least one was produced. We used the production of the plug as confirmation of pregnancy. In the case it was not produced after a week we separated the rats from each other and aimed to re-use them in another study. Female and male rats were placed in new freshly made-up cages upon separation of either successful or unsuccessful production of a vaginal plug. Then we waited for gestational day 12 where we either injected the pregnant rat with 2mg/kg or 5mg/kg of PolyI:C or a saline solution. The female rats' temperature was measured before the injection, 2,4, and 24 hours after the injection. The female rats' weight was also measured before injection and 24 hours after injection. We then waited for gestational day 17 where we started checking for births in the morning and afternoon. If a litter was born we marked the date on the cage card as post-natal day 0 (PND0). On PND7 we took one male and one female out of each litter to measure their heart rate and USVs. When taking the rats from the vivarium to the testing room we allowed them to

de-stress from the lighting change for 2 to 10 minutes before commencing measurement. When picking the pup out of the litter we either marked it with a capital 'I' for male or 'II' for female with a vivid marker. After picking a male or female pup, we would then start a timer for 15 minutes to ensure that the pups would not be separated from their mother for longer than this, as it could alter the behavior of the mother. The room we measured the pups in was at a steady temperature and only lit in red lighting when measuring. Recordings were between 8 to 12 minutes long depending on the quality of data observed at any point while measuring. After measurement, the pup was returned to the cage and a pup of the opposite sex would be picked, marked, and taken for measurement. This process would be repeated for the other litters that were born on the same day. On PND14 we then took the same male and female pup and repeated the measurements that were taken at PND7. Throughout the measurement phase, pups had to be remarked every day as the mother would groom the vivid marker completely off, if left alone for longer than a day. This way we were also able to monitor our rats' and pups' health by checking in on the litters every day.

Data Analysis

We only sought to analyse linear measurements of HRV, with RMSSD as a linear measurement of time and LF (% and nu), HF (% and nu), and LF/HF as linear measures of frequency. Once the ECG readings were collected, they were run through LabChart, a program which conducted a FFT analysis on the heartbeats, making different spectral bands accessible for frequency measures (HF and LF). For time measures, RMSSD was calculated by taking the root mean square of successive differences in NN intervals. The NN intervals were a product of RR intervals which were manually edited within LabChart by removing artefact (missed or premature beats).

We initially had 38 pups, however, due to editing of the HRV parameter data, we had to exclude two rats (both female, one administered saline and one 5mg/kg PolyI:C) from Cohort 2 PND14 as their RMSSD values were too high. We edited data in Excel by excluding rows with RMSSD values higher than 2.5. Every rat was included from Cohort 1. In the end we had 36 out of the initial 38 pups remaining for statistical testing. 11 saline pups (6 male, 5 female), 12 2mg/kg PolyI:C pups (6 male, 6 female), and 13 5mg/kg PolyI:C pups (7 male, 6 female). Medians were then calculated for the remaining individual pups' RMSSD, HF (%), HF (nu), LF (%), LF (nu), and LF/HF values. These medians were used to represent each pup in all six HRV measures in the repeated measures ANOVA. We conducted repeated measures ANOVA separately for each HRV measure.

Results

The independent variable was dose (saline, 2mg/kg PolyI:C, 5mg/kg PolyI:C). The dependent variables were RMSSD, LF (%), LF (nu), HF (%), HF (nu), and LF/HF. PND was a within subjects variable and dose and sex were between subjects variables.

Descriptives

RMSSD

For RMSSD, on average, there were no obvious sex differences between any of the doses or PNDs. However, regarding differences between PNDs, there were increases in RMSSD on average within all doses between PND7 and PND14, for example, saline male PND 7 ($M = 0.95$, $SD = 0.12$) and PND14 ($M = 1.30$, $SD = 0.41$), 2mg/kg PolyI:C male PND7 ($M = 0.88$, $SD = 0.15$) and PND14 ($M = 1.39$, $SD = 0.63$), and 5mg/kg PolyI:C male PND7 ($M = 1.02$, $SD = 0.13$) and PND14 ($M = 1.27$, $SD = 0.77$). This means that as the pups were maturing, their PNS was

increasing in influence over their ANS, and they were able to adapt to stressful situations more efficiently (see Figures 1 and 2) (Kidwell & Ellenbroek, 2018). More descriptive data on female RMSSD is displayed in Table A2.

Figure 1.

RMSSD of male and female pups, administered either saline, 2mg/kg of PolyI:C, or 5mg/kg of PolyI:C on PND7.

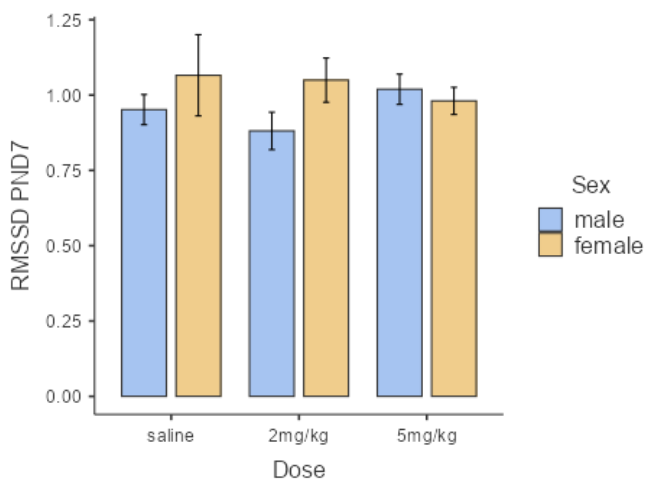
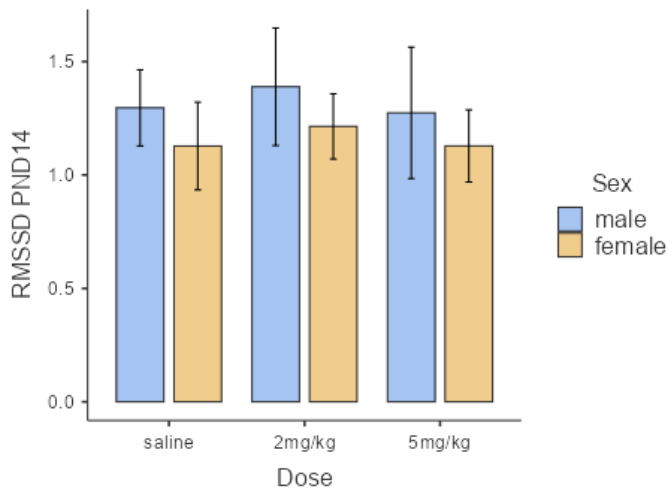


Figure 2.

RMSSD of male and female pups, administered either saline, 2mg/kg of PolyI:C, or 5mg/kg of PolyI:C on PND7.



LF (%)

For LF (%) , on average, there were no obvious sex differences between any of the doses or PNDs, except for the male ($M = 6.79$, $SD = 4.67$) and female ($M = 4.90$, $SD = 0.63$) saline pups on PND7. However, for saline males on PND7 the spread of the data is quite large, this could be indicative of more variation within this subgroup of our sample. Regarding a difference between PNDs, LF (%) increased between PND7 and PND14, for example, saline male PND 7 ($M = 6.79$, $SD = 4.67$) and PND14 ($M = 10.26$, $SD = 6.36$), 2mg/kg PolyI:C male PND7 ($M = 6.18$, $SD = 2.96$) and PND14 ($M = 10.51$, $SD = 12.99$), and 5mg/kg PolyI:C male PND7 ($M = 6.15$, $SD = 2.22$) and PND14 ($M = 12.72$, $SD = 5.36$). This means that the pups' SNS was more active at PND14 than on PND7 (EHRV, n. d.; Freitas et al., 2020; Kidwell & Ellenbroek, 2018). The only difference between doses, on average, was observed on PND14 between male ($M = 12.72$, $SD = 5.36$) and female 5mg/kg ($M = 10.97$, $SD = 9.57$) dose compared to saline, male (M

= 10.26, $SD = 6.36$) and female ($M = 9.30$, $SD = 6.18$), and 2mg/kg, male ($M = 10.51$, $SD = 12.99$) and female ($M = 14.53$, $SD = 17.73$). However, compared to RMSSD SD, looking at the SD of almost all LF (%) values, we cannot be confident about the mean as the spread of data appears to be unusual as visible in Figures 3 and 4. More descriptive data on LF (%) is displayed in Table B2.

Figure 3.

LF (%) of male and female pups, administered either saline, 2mg/kg of PolyI:C, or 5mg/kg of PolyI:C on PND7.

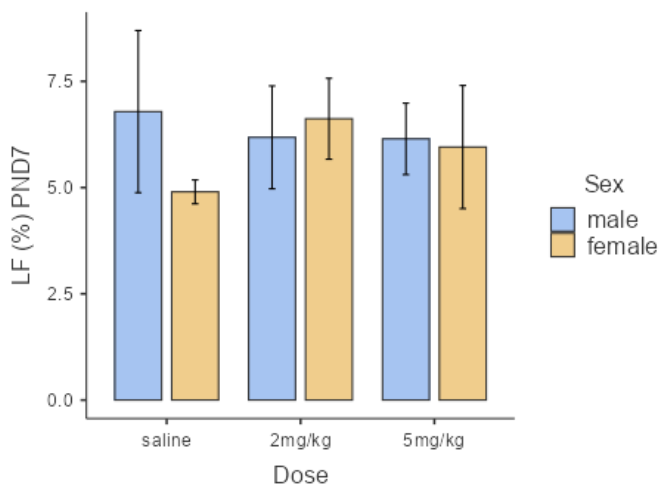
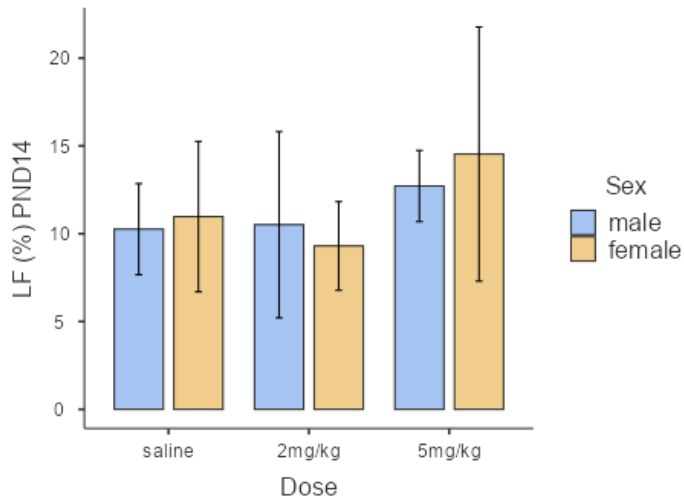


Figure 4.

LF (%) of male and female pups, administered either saline, 2mg/kg of PolyI:C, or 5mg/kg of PolyI:C on PND14.



LF (nu)

For LF (nu), on average, there were no obvious sex differences between any of the PNDs. However, there were dose differences on PND14, males who were administered 5mg/kg PolyI:C on PND14 ($M = 19.73$, $SD = 9.75$) compared to the females on PND14 ($M = 17.37$, $SD = 19.23$). This increase in LF (nu) at 5mg/kg PolyI:C also corresponds to the findings of LF (%) at 5mg/kg PolyI:C, where there was a difference between the 5mg/kg PolyI:C males and females on PND14 compared to the saline and 2mg/kg PolyI:C males and females on PND14. This means that on PND14 at 5mg/kg PolyI:C the SNS had increased activity compared to saline and 2mg/kg PolyI:C doses on PND14 and all doses on PND7 (EHRV, n. d.; Freitas et al., 2020; Kidwell & Ellenbroek, 2018). Regarding a difference between PNDs, an overall difference in LF (nu) was found between PND7 and PND14 across males and females, for example, saline male PND 7 ($M = 10.73$, $SD = 7.88$) and PND14 ($M = 14.37$, $SD = 10.55$), 2mg/kg PolyI:C male PND7 ($M =$

9.93, $SD = 5.41$) and PND14 ($M = 13.30$, $SD = 15.24$), and 5mg/kg PolyI:C male PND7 ($M = 7.88$, $SD = 3.69$) and PND14 ($M = 119.73$, $SD = 9.75$). However, compared to RMSSD SD, looking at the SD of almost all LF (nu) values, we cannot be confident about the mean as the spread of data appears to be unusual as visible in Figures 5 and 6. More descriptive data on LF (nu) is displayed in Table C2.

Figure 5.

LF (nu) of male and female pups, administered either saline, 2mg/kg of PolyI:C, or 5mg/kg of PolyI:C on PND7.

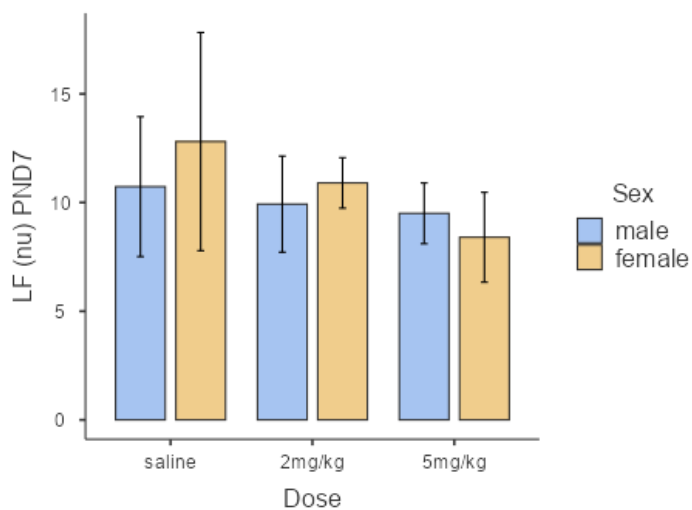
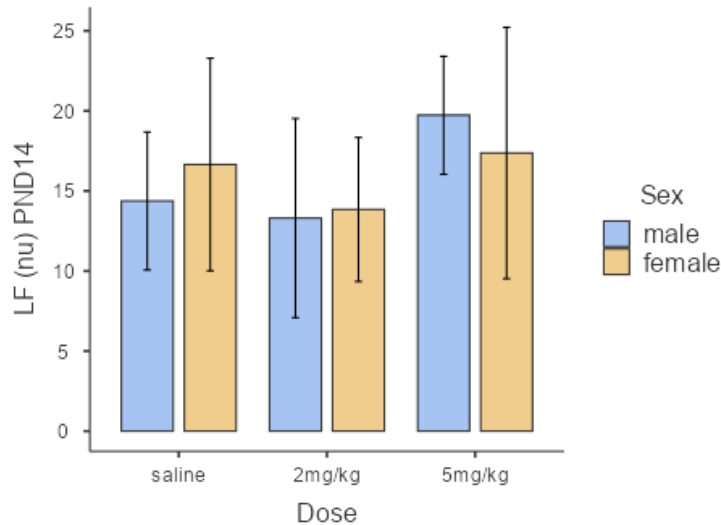


Figure 6.

LF (nu) of male and female pups, administered either saline, 2mg/kg of PolyI:C, or 5mg/kg of PolyI:C on PND14.



HF (%)

For HF (%), on average, there were no obvious sex differences between any of the doses or PNDs. Regarding dose differences, 2mg/kg PolyI:C males ($M = 49.37$, $SD = 9.31$) and females ($M = 52.18$, $SD = 18.64$) on PND14 had higher HF (%) than both saline PND14 males ($M = 45.50$, $SD = 8.54$) and females ($M = 41.03$, $SD = 7.94$) and 5mg/kg PolyI:C PND14 males ($M = 42.91$, $SD = 14.11$) and females ($M = 45.14$, $SD = 8.26$). Furthermore, the 2mg/kg PolyI:C group also had the largest increase of HF (%) from PND7 males ($M = 40.91$, $SD = 12.96$) and females ($M = 35.19$, $SD = 5.33$) to PND14 males ($M = 49.37$, $SD = 9.31$) and females ($M = 52.18$, $SD = 18.64$) compared to the other two dose groups (see Table D2). This means that for 2mg/kg PolyI:C the HF (%) increased which indicates an uptick in PNS activity (EHRV, n. d.; Freitas et al., 2020; Kidwell & Ellenbroek, 2018). However, compared to RMSSD SD, looking at

the SD of almost all HF (%) values, we cannot be confident about the mean, as the spread of data appears to be unusual as visible in Figures 7 and 8. More descriptive data on HF (%) is displayed in Table D2.

Figure 7.

HF (%) of male and female pups, administered either saline, 2mg/kg of PolyI:C, or 5mg/kg of PolyI:C on PND7.

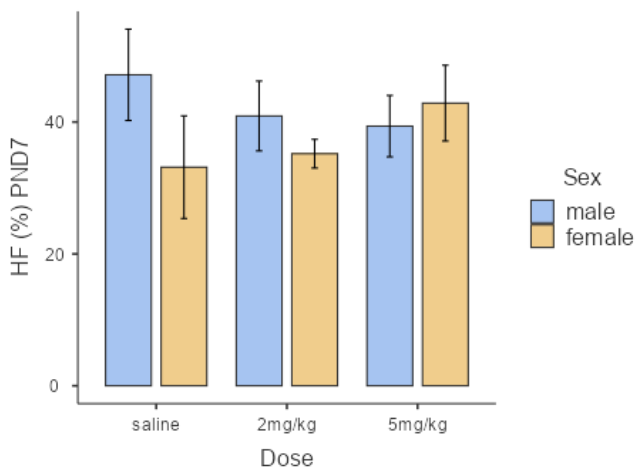
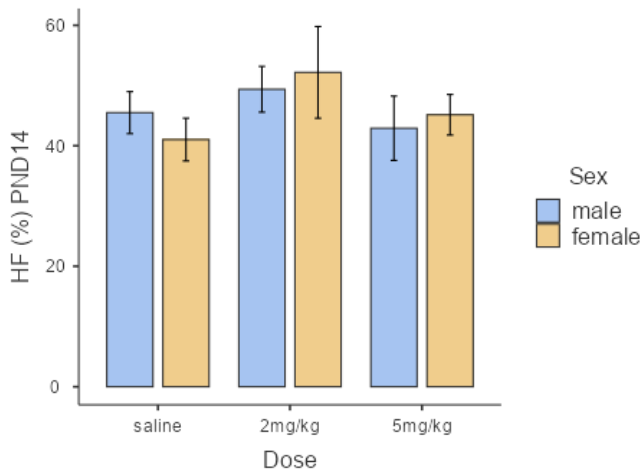


Figure 8.

HF (%) of male and female pups, administered either saline, 2mg/kg of PolyI:C, or 5mg/kg of PolyI:C on PND14.



HF (nu)

For HF (nu), there were slight sex differences as male HF (nu) was on average slightly higher than female HF (nu) in the saline, 2mg/kg PolyI:C, and 5mg/kg PolyI:C dose conditions (see Table E2), except for a slight difference in the 5mg/kg PolyI:C dose on PND14 where females ($M = 56.81$, $SD = 12.06$) had higher HF (nu) than males ($M = 56.43$, $SD = 8.68$). Regarding PND and dose differences looking at males and females in the saline condition, HF (nu) decreased from PND7 to PND14 for both males, from ($M = 63.13$, $SD = 12.15$) to ($M = 60.50$, $SD = 5.46$), and females, from ($M = 56.39$, $SD = 7.92$) to ($M = 54.76$, $SD = 1.91$). In the 5mg/kg PolyI:C dose condition, HF (nu) also decreased from PND7 to PND14 for males, from ($M = 59.85$, $SD = 12.07$) to ($M = 56.43$, $SD = 8.68$). However, HF (nu) remained almost identical for females in the 5mg/kg PolyI:C dose condition from PND7 to PND14, from ($M = 56.04$, $SD = 5.65$) to ($M = 56.81$, $SD = 12.06$). Strikingly, we saw an increase from PND7 to PND14 for both

males and females in the 2mg/kg PolyI:C dose condition; males' HF (nu) increased from ($M = 57.24$, $SD = 5.99$) to ($M = 68.92$, $SD = 6.12$), and females' HF (nu) increased from ($M = 54.07$, $SD = 10.91$) to ($M = 67.91$, $SD = 12.84$). Therefore, the saline and 5mg/kg PolyI:C dose conditions the pups' PNS activity decreased from PND7 to PND14, while in the 2mg/kg PolyI:C dose condition the PNS activity increased (as observed in the HF (%) measure) (EHRV, n. d.; Freitas et al., 2020; Kidwell & Ellenbroek, 2018). However, compared to RMSSD SD, looking at the SD of almost all HF (nu) values, we cannot be confident about the mean, as the spread of data appears to be unusual as visible in Figures 9 and 10.

Figure 9.

HF (nu) of male and female pups, administered either saline, 2mg/kg of PolyI:C, or 5mg/kg of PolyI:C on PND7.

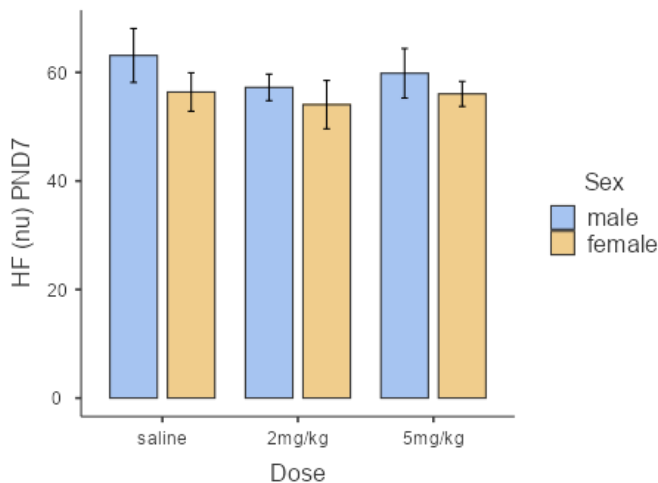
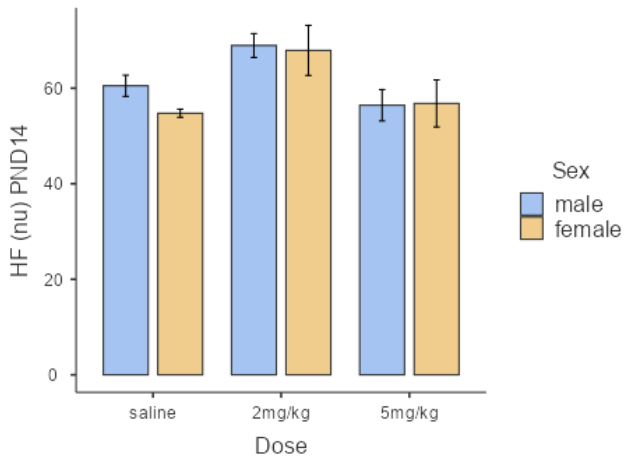


Figure 10.

HF (nu) of male and female pups, administered either saline, 2mg/kg of PolyI:C, or 5mg/kg of PolyI:C on PND14.



LF/HF

For LF/HF, there were slight sex differences, with the male pups attaining higher LF/HF values on average (see Table F2), except in the 5mg/kg dose condition where both sexes attained identical averages, but not SD, for each PND condition; males and females on PND7 attained ($M = 0.16$, $SD = 0.07$) and ($M = 0.16$, $SD = 0.11$) respectively, and on PND14 ($M = 0.35$, $SD = 0.22$) and ($M = 0.35$, $SD = 0.46$) respectively. Regarding PND and dose differences (see Figures 11 and 12), LF/HF increased from PND7 to PND14 within all dose conditions (see Table F2), with the smallest increase in the 2mg/kg PolyI:C dose condition: male PND7 ($M = 0.17$, $SD = 0.10$) to PND14 ($M = 0.20$, $SD = 0.23$), and female PND7 ($M = 0.19$, $SD = 0.05$) to PND14 ($M = 0.23$, $SD = 0.21$). The largest increase in LF/HF was observed in the 5mg/kg PolyI:C dose condition: male PND7 ($M = 0.16$, $SD = 0.07$) to PND14 ($M = 0.35$, $SD = 0.22$), and female PND7 ($M = 0.16$, $SD = 0.11$) to PND14 ($M = 0.35$, $SD = 0.46$). This means that 2mg/kg PolyI:C had the

largest effect on LF/HF, thus the balance between the PNS and SNS is more challenged in the 2mg/kg PolyI:C dose condition than in the saline and 5mg/kg PolyI:C dose conditions (Kidwell & Ellenbroek, 2018). We can be confident in our means for LF/HF measurements as they were small, meaning that most of our sample was close to the mean.

Figure 11.

LF/HF of male and female pups, administered either saline, 2mg/kg of PolyI:C, or 5mg/kg of PolyI:C on PND7.

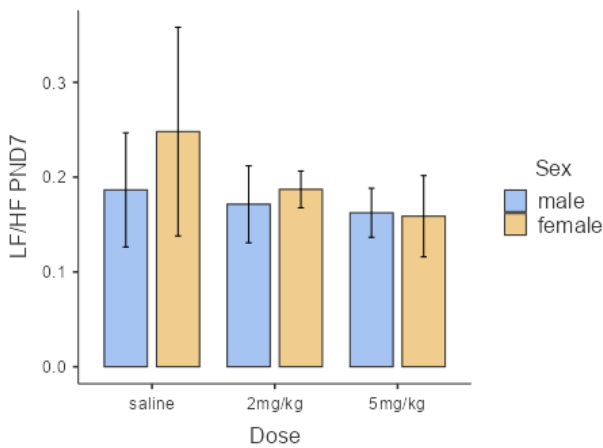
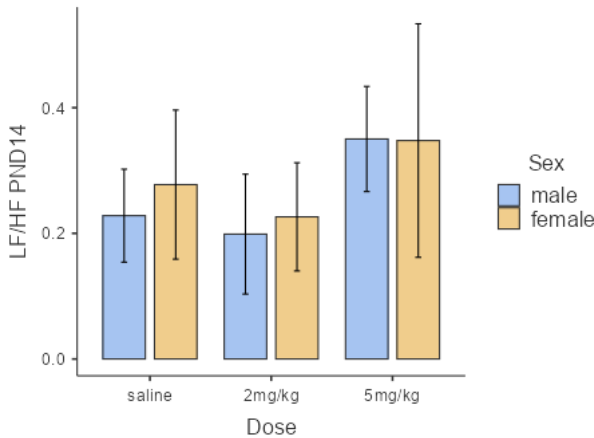


Figure 12.

LF/HF of male and female pups, administered either saline, 2mg/kg of PolyI:C, or 5mg/kg of PolyI:C on PND14.



Inferential Statistics

Six separate repeated measures ANOVA were performed to compare the effect of dose between PND7 and PND14 on the HRV measures RMSSD, LF (%), LF (nu), HF (%), HF (nu), and LF/HF (See Tables A1,B1, C1, D1, E1, and F1). Assumptions for Levene's were met for RMSSD on PND7($F = 1.22, p = .325$) and PND14 ($F = 0.49, p = .783$), LF (%) on PND7($F = 1.97, p = .112$) and PND14 ($F = 1.92, p = .121$), LF (nu) on PND7($F = 1.89, p = .127$) and PND14 ($F = 0.55, p = .737$), HF (%) on PND7($F = 1.23, p = .320$) and PND14 ($F = 2.04, p = .102$), and LF/HF on PND7($F = 2.24, p = .076$) and PND14 ($F = 0.70, p = .627$), and HF (nu) on PND7 ($F = 1.89, p = .126$) but not HF (nu) on PND14 ($F = 3.62, p = .011$). The violation of Levene's on PND14 may have been due to the small sample size within each PND and dose group. There were statistically significant differences between PND7 and PND14 in RMSSD ($F(1) = 6.1, p < .019$), LF (%) ($F(1) = 7.6, p < .010$), LF (nu) ($F(1) = 4.6, p < .039$), HF (%) ($F(1) = 4.3, p < .048$), and LF/HF ($F(1) = 2.7, p = .113$). However, the differences between PND7 and

PND14 were non-significant in HF (nu) ($F(1) = 6.10, p = .215$). No significant interactions were found between sexes. Only one statistically significant interaction was observed between PND and dose under the HRV parameter HF (nu) ($F(1) = 3.9, p < .031$). Therefore, we conducted a Tukey post hoc test (see Table E3). We found a significant correlation between 2mg/kg PolyI:C on PND14 and 5mg/kg PolyI:C on PND14 ($F(30) = 3.3, p < .026$). The means between 2mg/kg PolyI:C on PND14 male ($M = 68.92, SD = 6.12$) and female ($M = 67.91, SD = 12.84$), and 5mg/kg PolyI:C on PND14 male ($M = 56.43, SD = 8.68$) and female ($M = 56.81, SD = 12.06$), indicate that HF (nu) was higher in the 2mg/kg PolyI:C than 5mg/kg PolyI:C dose condition. This points to PNS activity decreasing as the dose of PolyI:C increased.

Discussion

Our study aim was to establish a relationship between prenatal infections and changes in HRV between offspring exposed to viral infections in utero vs those that were not. A potential relationship could encourage use of HRV measures as diagnostic tool of neurodevelopmental disorders in future research and treatment. We hypothesized that pups that were exposed to PolyI:C prenatally, would exhibit decreased HRV, whereas the pups whose mother was administered a saline solution during pregnancy would exhibit higher HRV compared to the PolyI:C pups.

HF (nu), one of the six parameters we used to measure HRV, produced a significant PND and dose interaction. The interaction which was confirmed by post hoc tests and a significant dose correlation was found between 2mg/kg PolyI:C on PND14 and 5mg/kg PolyI:C on PND14. Looking back at the means between 2mg/kg PolyI:C on PND14 and 5mg/kg PolyI:C on PND14, we found there to be higher HF (nu) in the 2mg/kg PolyI:C on PND14 compared to 5mg/kg PolyI:C on PND14. This indicates that there was higher PNS activity in the 2mg/kg PolyI:C dose

condition. Higher PNS activity is indicative of higher acetylcholine activity, the neurotransmitter that works to inhibit the heart muscles, slowing down the heart rate and increasing activity of the vagus nerve (Kemp & Quintana, 2013; Kidwell & Ellenbroek, 2018). An increase in vagus nerve activity/vagal tone is indicative of an increased ability to relax after a stressful situation.

Therefore, the pups' exposed to 2mg/kg PolyI:C on PND14 are more able to relax after stressful situations than the 5mg/kg PolyI:C pups on PND14. This finding, however, appears less reliable when considering the violation of assumptions, only for PND14, in the repeated measures ANOVA of the HF (nu) HRV parameter. One can speculate this is because of the small sample sizes within each dose and PND group: saline (N = 11), 2mg/kg PolyI:C (N = 12), and 5mg/kg PolyI:C (N = 13) for both PNDs. When looking at the tendencies between doses on PND14, this trend does not hold up for all HRV parameters when looking at Figures (2, 4, 6, 8, 10, and 12). RMSSD (Figure 2), HF (%) (Figure 8), and HF (nu) (Figure 10) show a tendency of the 2mg/kg dose being higher in their respective HRV parameter than the 5mg/kg dose, whereas LF (%) (Figure 4), LF (nu) (Figure 6), and LF/HF (Figure 12) show a tendency of the 2mg/kg dose being lower in their respective HRV parameter than the 5mg/kg dose. This consistency in the lower PolyI:C dose having higher values being consistent over HF measures is logical as HF is affected more by the PNS than SNS (as LF measures are) (Kidwell & Ellenbroek, 2018).

Regarding observable tendencies between PND7 and PND14 across all 6 parameters, some interesting trends emerged. Looking at RMSSD, the 2mg/kg dose condition had lower values than saline and 5mg/kg on PND7, however the reverse was true for PND14. It appears that over the pups' development, those exposed to 2mg/kg PolyI:C rather than saline or the 5mg/kg, ended up experiencing higher PNS activity, this was also observed under HF (%) and HF (nu) (Figures 1-2, 7-8, and 9-10). Looking at the remaining 3 parameters, for example under

LF/HF values decreased from PND7 to PND14 in the 2mg/kg dose condition, this was also observed under LF (%) and LF (nu) (Figures 3-4, 5-6, and 11-12). Thus, it is implied that in the 2mg/kg dose condition there was less SNS activity over time compared to the saline and 5mg/kg dose conditions. This balance between RMSSD/HF (%) /HF (nu) and LF (%) /LF (nu) /LF/HF corresponds to existing literature on HRV and its underlying PNS and SNS balance ((Kemp & Quintana, 2013; Kidwell & Ellenbroek, 2018). What is intriguing is the fact that the 2mg/kg dose showed the most PNS and SNS activity on PND14 compared to saline and even the higher 5mg/kg dose of PolyI:C. Why did the higher dose not have a greater effect on PNS and SNS? Murray et al. (2019) conducted a study that proves that a dose of 5mg/kg PolyI:C or even at 10mg/kg does have an influence on the dam's neuroimmune system, as they found increased neuroinflammation through the increased presence of microglia in their PolyI:C treated dams compared to saline. Therefore, our 5mg/kg dose should have produced equal or even greater results to our 2mg/kg dose. In a mice model focusing on the effect of differing viral loads on HBV infected mice, Dusseaux et al. (2017) found there to be more T-cells present in the animal's liver the higher the viral load. Therefore, the 5mg/kg dose in our study may have been processed more efficiently by the dam's immune system due to the higher viral load than the potentially lower viral load at a 2mg/kg dose. Alternatively, there may have been something wrong with the 5mg/kg PolyI:C batch used to inject the rats, as it produced similar results to the saline solution.

Applications

The 2mg/kg dose that appeared to have produced the most change in HRV across all six measured parameters, assuming this is true, this finding may encourage future research into looking at rather lower doses of PolyI:C compared to higher doses. On the other hand, assuming

that the 5mg/kg batch of PolyI:C may have been defective, it would be wise to repeat this experiment with an even closer eye on the process of manufacture.

The observed tendencies of PolyI:C (from Figures 1 – 12) altering HRV, from PND7 to PND14, are evidence that HRV is a useful measure of changes induced through prenatal viral infections. This will encourage future researchers interested in prenatal viral infections to use HRV measures, which will ultimately lead to the development of the theory, technology, and methods used.

Limitations

Our design was of a short experimental nature, this may have limited the scope to which one could apply the HRV data to the development of a rat. We only measured twice within the first two weeks of the pups' lives; this means we may have missed key developmental windows that may be influenced by prenatal viral infections. Furthermore, we only recorded the pups' HRV over an 8- to 12-minute window. Over the course of 24 hours, a rat's HRV usually fluctuates between being awake and sleeping (Rowan et al., 2007). Therefore, future studies may want to differentiate between HRV recorded during wake and sleep. Additionally, rats' HRV is affected differently by respiration than larger mammals, being influenced more during inspiration (Rowan et al., 2007). Therefore, results from this study may not be as transferable to human physiology, as there are differing biological mechanisms at play between the two species.

The sample size, in total, and in the different dose and sex groups, may have been too small to capture of HRV activity in rats. There is limited research on HRV measures applied to rodent models, so we do not exactly know how many individuals are required by group to attain reliable results. Usually, 8 – 10 individuals within a group are sufficient in rodent models, but

this has not yet been determined for rodent models measuring HRV. Due to the small sample size, we also obtained large SD for all measures, except RMSSD and LF/HF. This may have also been due to the well-established theory and use of RMSSD as a reliable and valid measure of HRV. However, this is not the case with LF/HF.

The variation in spread (SD) from the mean appeared higher on PND14 compared to PND7 across all HRV parameters, even within RMSSD and LF/HF. This may have been due to the testing equipment used. At PND14 the rat pups were significantly bigger, stronger, and more mobile than they were at PND7. While recording on PND14, it was not uncommon that the pup would escape the ECGenie and recordings would have to start over or be impacted due to the stress of placing the pup back into the ECGenie. We overcame this by fastening the metal cage with bands for all PND14 recordings after we had noticed this trend. However, the pups were still moving more inside the ECGenie, took longer to calm down, and partook in more grooming behaviours during recording. Therefore, the values acquired on PND14 may have differed more between each other due to the more active nature of the older pups.

According to Kidwell and Ellenbroek (2018) the HRV parameter LF/HF operates under four assumptions which are not currently met in the literature and require further assessment to be reliably used as a measure of HRV. Therefore, the tendencies described regarding the LF/HF parameter may not be as reliable due to the understudied nature of the parameter. Therefore, future studies should be cautious when applying this parameter and perhaps conduct further research specifically into the mechanisms of HRV's time and frequency domain measures.

We limited our explanatory power by not measuring the dam's HRV before and during pregnancy. Therefore, we were not able to check for potential heritable effects on HRV. This may have been more important than in other rodent studies as our sample size was quite small.

Furthermore, we did not measure a baseline HRV for an untreated pup from a normal pregnancy. Our control group was only injected with saline; however, it would have been helpful to compare HRV between pups whose mother did not go through the unavoidable stresses of this experiment and those in the control group to get a sense of a baseline HRV for Sprague Dawley pups.

Conclusion

This study aimed to establish a relationship between prenatal infections and changes in HRV between offspring exposed to viral infections in utero vs those that were not. We hypothesized that pups exposed to PolyI:C vs a saline solution, during gestation, will have decreased HRV, whereas pups whose mother was administered the saline solution will see a higher HRV than the PolyI:C pups. We found one significant PND and dose interaction within the HF (nu) HRV parameter, and upon further examination found that the differences lay between the 2mg/kg and 5mg/kg PolyI:C doses, with the 2mg/kg dose producing higher HF (nu) values. This meant that pups in the 2mg/kg dose condition were experiencing higher PNS activity than those in the 5mg/kg dose condition, meaning that the pups' HRV of the lower dose of PolyI:C was affected more than those who experienced the higher dose. Taking a closer look at the tendencies between the PND and dose means, we noticed a similar trend as demonstrated by the PND and dose interaction, where 2mg/kg dose was indicative of greater changes in HRV than the higher dose. However, these results should be interpreted cautiously as our study design did not account for wake/sleep changes in HRV, did not have a baseline HRV of dam or pup to compare the current study's results to, had a small sample size per group, and due to the under-researched nature of some of the applied HRV parameters. Nevertheless, it is important to keep in mind that this was a pilot study and the first of its kind, therefore, methodological, and theoretical challenges are to be expected from such a naïve field.

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Appendix A

RMSSD Repeated Measures ANOVA and Descriptives Tables

Table A1.

Repeated Measures ANOVA RMSSD PND7 vs PND14 and Assumptions

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p
PND	1.09	1	1.09	6.10	0.019
PND * Sex	0.27	1	0.27	1.49	0.231
PND * Dose	0.07	2	0.04	0.20	0.818
PND * Sex * Dose	0.05	2	0.02	0.13	0.877
Residual	5.34	30	0.18		

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p
Sex	0.03	1	0.03	0.22	0.643
Dose	0.01	2	0.01	0.05	0.949
Sex * Dose	0.03	2	0.01	0.10	0.907
Residual	4.06	30	0.14		

Note. Type 3 Sums of Squares

Assumptions

Homogeneity of Variances Test (Levene's)

	F	df1	df2	p
RMSSD PND7	1.22	5	30	0.325
RMSSD PND14	0.49	5	30	0.783

Table A2.*RMSSD Descriptives PND7 vs PND14*

	Dose	Sex	RMSSD PND7	RMSSD PND14
N	saline	male	6	6
		female	5	5
	2mg/kg	male	6	6
		female	6	6
	5mg/kg	male	7	7
		female	6	6
Mean	saline	male	0.95	1.30
		female	1.07	1.13
	2mg/kg	male	0.88	1.39
		female	1.05	1.21
	5mg/kg	male	1.02	1.27
		female	0.98	1.13
Median	saline	male	0.97	1.29
		female	0.94	0.98

	Dose	Sex	RMSSD PND7	RMSSD PND14
	2mg/kg	male	0.82	1.16
		female	1.00	1.10
	5mg/kg	male	1.00	1.00
		female	0.98	1.02
Standard deviation	saline	male	0.12	0.41
		female	0.30	0.43
	2mg/kg	male	0.15	0.63
		female	0.18	0.35
	5mg/kg	male	0.13	0.77
		female	0.11	0.39

Appendix B

LF (%) Repeated Measures ANOVA and Descriptives Tables

Table B1.

Repeated Measures ANOVA LF (%) PND7 vs PND14

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p
PND	498.14	1	498.14	7.64	0.010
PND * Sex	4.35	1	4.35	0.07	0.798
PND * Dose	54.21	2	27.10	0.42	0.664
PND * Sex * Dose	15.64	2	7.82	0.12	0.887
Residual	1956.46	30	65.22		

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p
Sex	0.05	1	0.05	0.00	0.976
Dose	44.92	2	22.46	0.40	0.671
Sex * Dose	7.04	2	3.52	0.06	0.939
Residual	1665.68	30	55.52		

Note. Type 3 Sums of Squares

Assumptions

Homogeneity of Variances Test (Levene's)

	F	df1	df2	p
LF (%) PND7	1.97	5	30	0.112
LF (%) PND14	1.92	5	30	0.121

Table B2.*LF (%) Descriptives PND7 vs PND14*

Descriptives

	Dose	Sex	LF (%) PND7	LF (%) PND14
N	saline	male	6	6
		female	5	5
	2mg/kg	male	6	6
		female	6	6
	5mg/kg	male	7	7
		female	6	6
Mean	saline	male	6.79	10.26
		female	4.90	10.97
	2mg/kg	male	6.18	10.51
		female	6.62	9.30
	5mg/kg	male	6.15	12.72
		female	5.95	14.53

Descriptives

	Dose	Sex	LF (%) PND7	LF (%) PND14
Median	saline	male	5.39	11.34
		female	4.66	8.86
	2mg/kg	male	5.91	2.62
		female	5.81	8.63
	5mg/kg	male	5.95	14.97
		female	6.24	5.79
Standard deviation	saline	male	4.67	6.36
		female	0.63	9.57
	2mg/kg	male	2.96	12.99
		female	2.33	6.18
	5mg/kg	male	2.22	5.36
		female	3.55	17.73

Appendix C

LF (nu) Repeated Measures ANOVA and Descriptives Tables

Table C1.

Repeated Measures ANOVA LF (nu) PND7 vs PND14

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p
PND	539.53	1	539.53	4.64	0.039
PND * Sex	1.09	1	1.09	0.01	0.923
PND * Dose	157.64	2	78.82	0.68	0.515
PND * Sex * Dose	1.60	2	0.80	0.01	0.993
Residual	3486.56	30	116.22		

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p
Sex	2.88	1	2.88	0.03	0.874
Dose	46.56	2	23.28	0.21	0.813
Sex * Dose	47.01	2	23.51	0.21	0.811
Residual	3349.47	30	111.65		

Note. Type 3 Sums of Squares

Assumptions

Homogeneity of Variances Test (Levene's)

	F	df1	df2	p
LF (nu) PND7	1.89	5	30	0.127
LF (nu) PND14	0.55	5	30	0.737

Table C2.*LF (nu) Descriptives PND7 vs PND14*

Descriptives

	Dose	Sex	LF (nu) PND7	LF (nu) PND14
N	saline	male	6	6
		female	5	5
	2mg/kg	male	6	6
		female	6	6
	5mg/kg	male	7	7
		female	6	6
Mean	saline	male	10.73	14.37
		female	12.80	16.65
	2mg/kg	male	9.93	13.30
		female	10.90	13.84
	5mg/kg	male	9.50	19.73
		female	8.40	17.37

Descriptives

	Dose	Sex	LF (nu) PND7	LF (nu) PND14
Median	saline	male	9.89	13.90
		female	7.29	11.21
	2mg/kg	male	10.33	4.52
		female	11.66	11.19
	5mg/kg	male	8.61	19.69
		female	9.59	9.49
Standard deviation	saline	male	7.88	10.55
		female	11.22	14.84
	2mg/kg	male	5.41	15.24
		female	2.85	11.02
	5mg/kg	male	3.69	9.75
		female	5.06	19.23

Appendix D
HF (%) Repeated Measures ANOVA and Descriptives Tables

Table D1.

Repeated Measures ANOVA HF (%) PND7 vs PND14

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p
PND	696.01	1	696.01	4.25	0.048
PND * Sex	139.91	1	139.91	0.86	0.362
PND * Dose	377.67	2	188.83	1.15	0.329
PND * Sex * Dose	109.41	2	54.70	0.33	0.718
Residual	4908.19	30	163.61		

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p
Sex	121.47	1	121.47	0.74	0.398
Dose	88.83	2	44.42	0.27	0.766
Sex * Dose	439.19	2	219.59	1.33	0.279
Residual	4946.33	30	164.88		

Note. Type 3 Sums of Squares

Assumptions

Homogeneity of Variances Test (Levene's)

	F	df1	df2	p
HF (%) PND7	1.23	5	30	0.320
HF (%) PND14	2.04	5	30	0.102

Table D2.*HF (%) Descriptives PND7 vs PND14*

Descriptives

	Dose	Sex	HF (%) PND7	HF (%) PND14
N	saline	male	6	6
		female	5	5
	2mg/kg	male	6	6
		female	6	6
	5mg/kg	male	7	7
		female	6	6
Mean	saline	male	47.16	45.50
		female	33.15	41.03
	2mg/kg	male	40.91	49.37
		female	35.19	52.18
	5mg/kg	male	39.37	42.91
		female	42.86	45.14

Descriptives

	Dose	Sex	HF (%) PND7	HF (%) PND14
Median	saline	male	38.39	45.44
		female	34.40	39.17
	2mg/kg	male	38.58	50.91
		female	32.85	52.75
	5mg/kg	male	36.74	44.07
		female	46.90	41.94
Standard deviation	saline	male	16.96	8.54
		female	17.40	7.94
	2mg/kg	male	12.96	9.31
		female	5.33	18.64
	5mg/kg	male	12.31	14.11
		female	14.08	8.26

Appendix E
HF (nu) Repeated Measures ANOVA and Descriptives Tables

Table E1.

Repeated Measures ANOVA HF (nu) PND7 vs PND14

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p
PND	171.22	1	171.22	1.61	0.215
PND * Sex	26.68	1	26.68	0.25	0.620
PND * Dose	835.65	2	417.82	3.92	0.031
PND * Sex * Dose	7.86	2	3.93	0.04	0.964
Residual	3194.58	30	106.49		

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p
Sex	199.90	1	199.90	3.06	0.090
Dose	293.06	2	146.53	2.25	0.123
Sex * Dose	71.97	2	35.98	0.55	0.582
Residual	1957.20	30	65.24		

Note. Type 3 Sums of Squares

Assumptions

Homogeneity of Variances Test (Levene's)

	F	df1	df2	p
HF (nu) PND7	1.89	5	30	0.126
HF (nu) PND14	3.62	5	30	0.011

Table E2.*HF (nu) Descriptives PND7 vs PND14*

Descriptives

	Dose	Sex	HF (nu) PND7	HF (nu) PND14
N	saline	male	6	6
		female	5	5
	2mg/kg	male	6	6
		female	6	6
	5mg/kg	male	7	7
		female	6	6
Mean	saline	male	63.13	60.50
		female	56.39	54.76
	2mg/kg	male	57.24	68.92
		female	54.07	67.91
	5mg/kg	male	59.85	56.43
		female	56.04	56.81

Descriptives

	Dose	Sex	HF (nu) PND7	HF (nu) PND14
Median	saline	male	61.24	63.03
		female	53.42	55.96
	2mg/kg	male	56.96	68.55
		female	55.24	70.89
	5mg/kg	male	61.93	61.47
		female	56.15	54.82
Standard deviation	saline	male	12.15	5.46
		female	7.92	1.91
	2mg/kg	male	5.99	6.12
		female	10.91	12.84
	5mg/kg	male	12.07	8.68
		female	5.65	12.06

Table E3.*HF (nu) Post Hoc comparisons of PND and Dose interactions*

Comparison									
PND	Dose		PND	Dose	Mean Difference	SE	df	t	p_{Tukey}
PND7	saline	-	PND7	2mg/kg	4.10	4.04	30.00	1.02	0.909
		-	PND7	5mg/kg	1.82	3.97	30.00	0.46	0.997
		-	PND14	saline	2.13	4.42	30.00	0.48	0.996
		-	PND14	2mg/kg	-8.65	3.88	30.00	-2.23	0.255
		-	PND14	5mg/kg	3.14	3.82	30.00	0.82	0.961
	2mg/kg	-	PND7	5mg/kg	-2.29	3.87	30.00	-0.59	0.991
		-	PND14	saline	-1.97	3.87	30.00	-0.51	0.995
		-	PND14	2mg/kg	-12.76	4.21	30.00	-3.03	0.052
		-	PND14	5mg/kg	-0.96	3.72	30.00	-0.26	1.000
		-	PND14	5mg/kg	0.32	3.80	30.00	0.08	1.000
	5mg/kg	-	PND14	2mg/kg	-10.47	3.71	30.00	-2.82	0.081
		-	PND14	5mg/kg	1.33	4.06	30.00	0.33	0.999
PND14	saline	-	PND14	2mg/kg	-10.79	3.71	30.00	-2.91	0.067
		-	PND14	5mg/kg	1.01	3.64	30.00	0.28	1.000
	2mg/kg	-	PND14	5mg/kg	11.80	3.55	30.00	3.32	0.026

Appendix F
LF/HF Repeated Measures ANOVA and Descriptives Tables

Table F1.

Repeated Measures ANOVA LF/HF PND7 vs PND14

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p
PND	0.13	1	0.13	2.66	0.113
PND * Sex	0.00	1	0.00	0.00	0.998
PND * Dose	0.10	2	0.05	0.99	0.382
PND * Sex * Dose	0.00	2	0.00	0.00	0.996
Residual	1.48	30	0.05		

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p
Sex	0.01	1	0.01	0.25	0.619
Dose	0.04	2	0.02	0.52	0.602
Sex * Dose	0.01	2	0.01	0.12	0.889
Residual	1.29	30	0.04		

Note. Type 3 Sums of Squares

Assumptions

Homogeneity of Variances Test (Levene's)

	F	df1	df2	p
LF/HF PND7	2.24	5	30	0.076
LF/HF PND14	0.70	5	30	0.627

Table F2.*LF/HF Descriptives PND7 vs PND14*

Descriptives

	Dose	Sex	LF/HF PND7	LF/HF PND14
N	saline	male	6	6
		female	5	5
	2mg/kg	male	6	6
		female	6	6
	5mg/kg	male	7	7
		female	6	6
Mean	saline	male	0.19	0.23
		female	0.25	0.28
	2mg/kg	male	0.17	0.20
		female	0.19	0.23
	5mg/kg	male	0.16	0.35
		female	0.16	0.35
Median	saline	male	0.18	0.19

Descriptives

	Dose	Sex	LF/HF PND7	LF/HF PND14
		female	0.12	0.16
	2mg/kg	male	0.18	0.06
		female	0.18	0.16
	5mg/kg	male	0.15	0.30
		female	0.17	0.15
Standard deviation	saline	male	0.15	0.18
		female	0.25	0.27
	2mg/kg	male	0.10	0.23
		female	0.05	0.21
	5mg/kg	male	0.07	0.22
		female	0.11	0.46